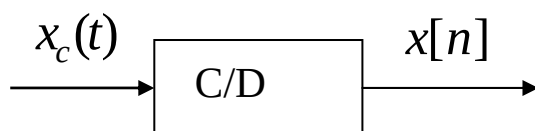


Chapter 4 Sampling of Continuous-time Signals

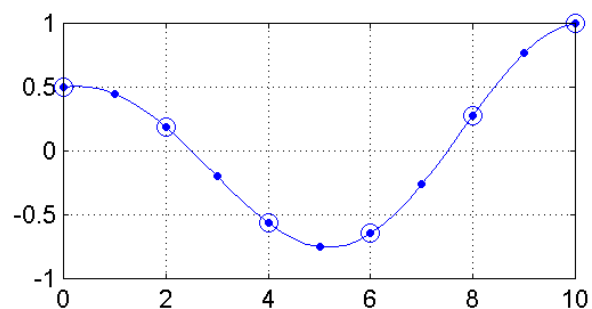
- Advantages of digital signal processing, e.g., audio/video CD.
- Things to look at:
 - Continuous-to-discrete (C/D)
 - Discrete-to-continuous (D/C) – perfect reconstruction
 - Frequency-domain analysis of sampling process
 - Sampling rate conversion

4.1 Periodic Sampling

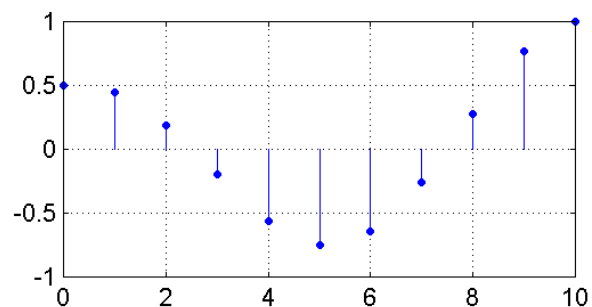
- Ideal continuous-to-discrete-time (C/D) converter



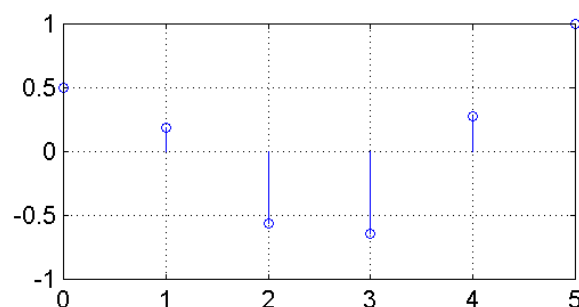
$x_c(t)$



$T=1, x[n] = x_c(nT)$



$T=2, x[n] = x_c(nT)$



Continuous-time signal: $x_c(t)$

Discrete-time signal: $x[n] = x_c(nT)$, $-\infty < n < \infty$, T : sampling period

How to model this C/D process?

In theory, we break the C/D operation into two steps:

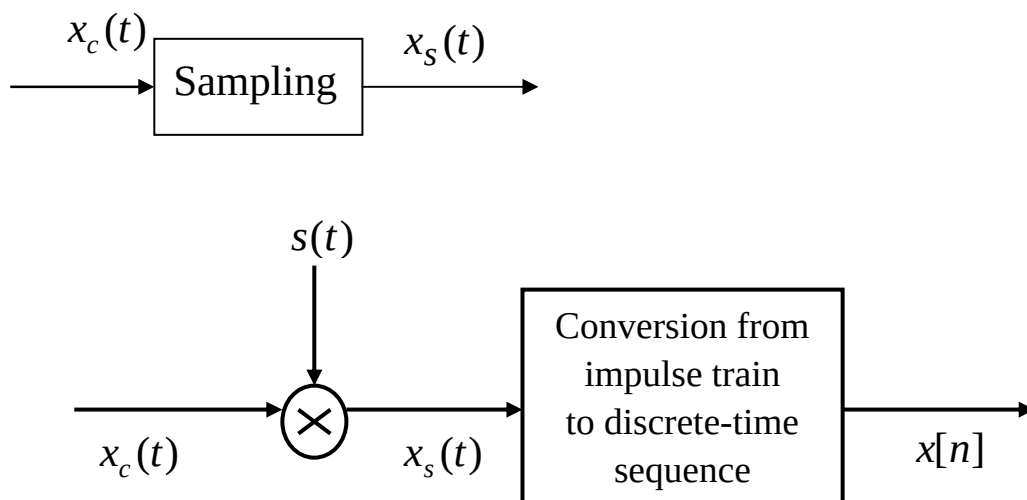
- (1) Ideal sampling using “analog delta function (impulse)”
- (2) Conversion from impulse train to discrete-time sequence

Step (1) can be modeled by mathematical equation.

Step (2) is a “concept”, no mathematical model.

In reality, the electronic analog-to-digital (A/D) circuits can approximate the ideal C/D operation. This circuitry is one piece; it cannot be split into two steps.

● Ideal sampling



Ideal sampling signal: impulse train (an analog signal)

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT), \quad T: \text{sampling period}$$

Analog (continuous-time) signal: $x_c(t)$

Sampled (continuous-time) signal: $x_s(t)$

$$\begin{aligned} x_s(t) &= x_c(t)s(t) = x_c(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) \\ &= \sum_{n=-\infty}^{\infty} x_c(t) \delta(t - nT) = \sum_{n=-\infty}^{\infty} x_c(nT) \delta(t - nT) \end{aligned}$$

4.2 Frequency-domain Representation of Sampling

$$s(t) \leftrightarrow S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s) \quad , \text{ where } \Omega_s = 2\pi / T$$

Remark: Ω : analog frequency (radians/s)

ω : discrete (normalized) frequency (radians/sample)

$$\Omega = \omega/T ; \quad -\pi < \omega \leq \pi , \quad -\frac{\pi}{T} < \Omega \leq \frac{\pi}{T}$$

Step 1: *Ideal Sampling* (all in analog domain)

$$\begin{aligned} X_s(j\Omega) &= \frac{1}{2\pi} X_c(j\Omega) * S(j\Omega) = \frac{1}{T} X_c(j\Omega) * \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s) \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j\Omega) * \delta(\Omega - k\Omega_s) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s)) \end{aligned}$$

The sampled signal spectrum is the sum of shifted copies of the original.

Remark: In analog domain,

$$\begin{aligned} x(t)y(t) &\leftrightarrow X(f) * Y(f) \\ &= \frac{1}{2\pi} X(j\Omega) * Y(j\Omega) \end{aligned}$$

Step 2: *Analog Impulses to Sequence* (analog to discrete-time)

No mathematical model. The spectrum of $x_s(t)$, $X_s(j\Omega)$, is the

same as the spectrum of $x[n]$, $X(e^{j\Omega T})$. (See the Appendix at the end.)

$$\text{Now, } X(e^{j\Omega T}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

$$\text{Thus, } X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$$

Remark: In time domain, $x_s(t)$ and $x[n]$ are two very different signals but they have the “same” spectra in frequency domain.

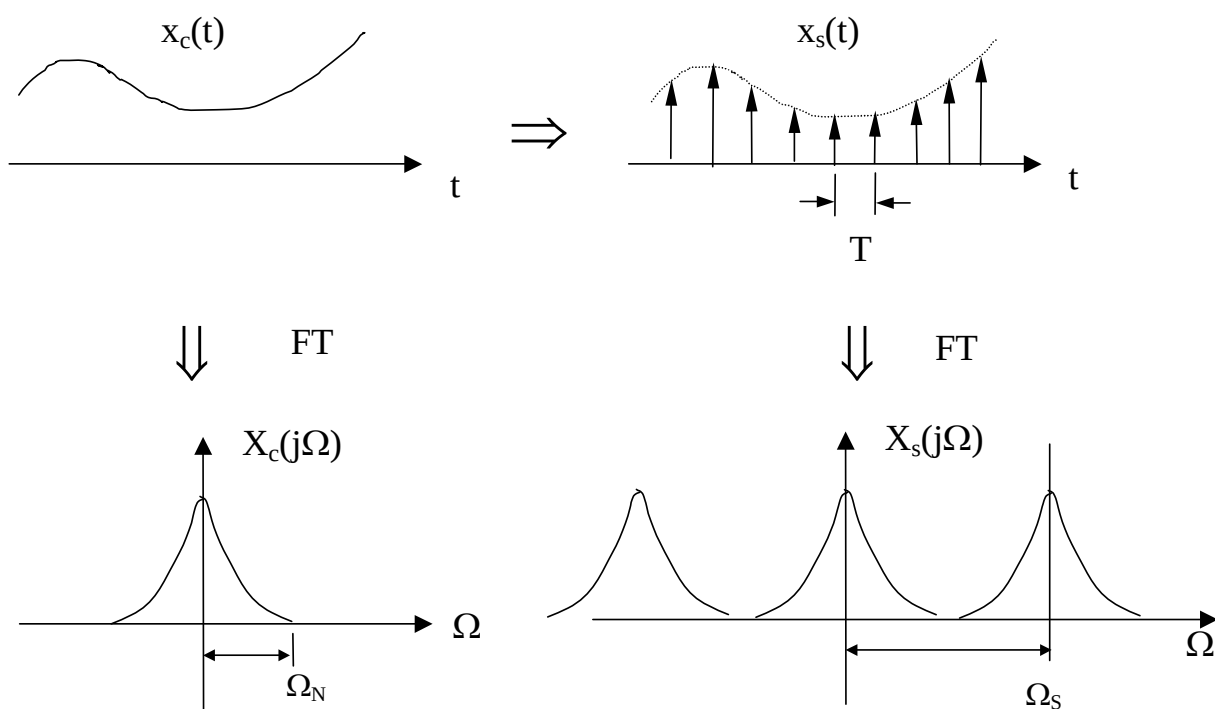
Two Cases:

(1) no aliasing: $\Omega_s > 2\Omega_N$, and

(2) aliasing: $\Omega_s < 2\Omega_N$, where Ω_N is the highest nonzero frequency component of $X_c(j\Omega)$.

After sampling, the replicas of $X_c(j\Omega)$ overlap (in frequency domain). That is, the higher frequency components of $X_c(j\Omega)$ overlap with the lower frequency components of $X_c(j(\Omega - \Omega_s))$.

(Fig. 4.3 on O&S, p.186)



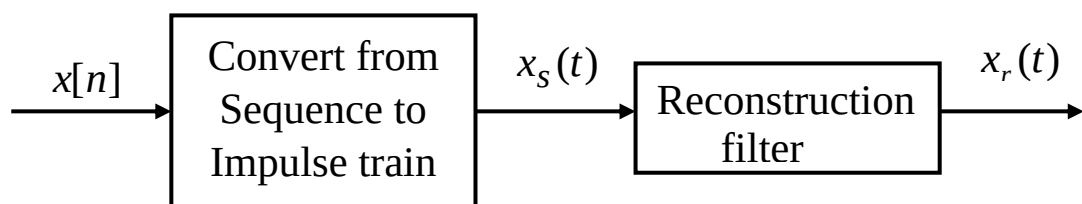
Nyquist Sampling Theorem: Let $x(t)$ be a *bandlimited* signal with $X_c(j\Omega) = 0$ for $|\Omega| \geq \Omega_N$. (i.e., no components at frequencies greater than Ω_N) Then $x_c(t)$ is uniquely determined by its samples

$$x[n] = x_c(nT), n = 0, \pm 1, \pm 2, \dots, \text{ if } \Omega_s = \frac{2\pi}{T} \geq 2\Omega_N. \text{ (Nyquist, Shannon)}$$

- **Nyquist frequency** $= \Omega_N$, the bandwidth of signal.
- **Nyquist rate** $= 2\Omega_N$, the minimum sampling rate without distortion. (In some books, Nyquist frequency = Nyquist rate.)
- **Undersampling:** $\Omega_s < 2\Omega_N$
- **Oversampling:** $\Omega_s > 2\Omega_N$

4.3 Reconstruction of a Bandlimited Signal from Its Samples

-- **Perfect reconstruction:** recover the original continuous-time signal without distortion, e.g., ideal lowpass (bandpass) filter



Based on the frequency-domain analysis, if we can “clip” one copy of the original spectrum, $X_c(j\Omega)$, without distortion, we can achieve the perfect reconstruction. For example, we use the ideal low-pass filter as the reconstruction filter.

Remark: Note that $x_s(t)$ is an analog signal (impulses).

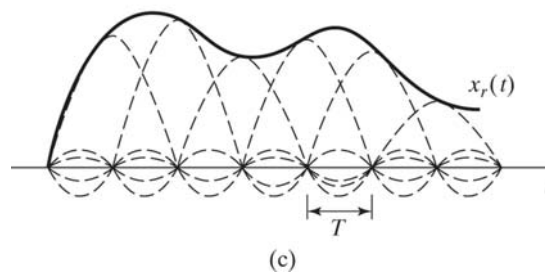
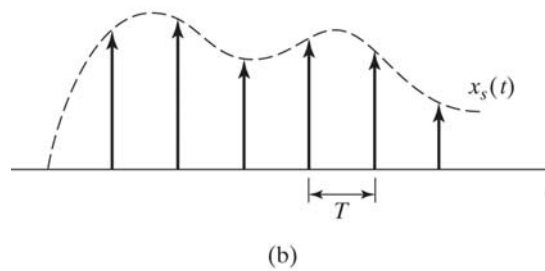
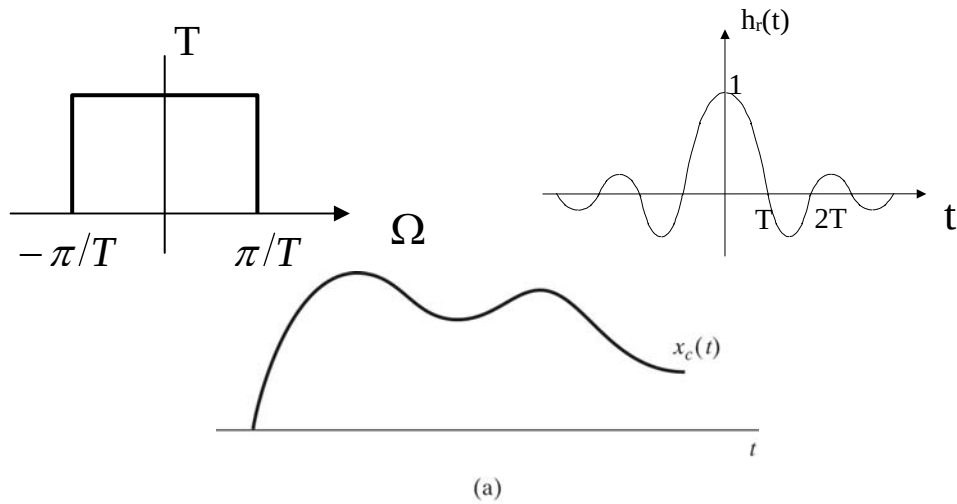
$$x_c(t) \rightarrow \text{sampling} \rightarrow x_s(t) = \sum x(nT)\delta(t - nT) \rightarrow \text{seq. - convr.} \rightarrow x[n]$$

$$x[n] \rightarrow \text{impulse-convr.} \rightarrow x_s(t) = \sum x[n]\delta(t - nT) \rightarrow \text{recon.} \rightarrow x_r(t)$$

$$\begin{aligned} x_r(t) &= x_s(t) * h_r(t) = \int_{-\infty}^{\infty} \left\{ \sum_{n=-\infty}^{\infty} x[n]\delta(\lambda - nT)h_r(t - \lambda) \right\} d\lambda \\ &= \sum_{n=-\infty}^{\infty} \left\{ x[n] \int_{-\infty}^{\infty} \delta(\lambda - nT)h_r(t - \lambda) d\lambda \right\} = \sum_{n=-\infty}^{\infty} x[n]h_r(t - nT) \end{aligned}$$

Ideal low-pass reconstruction filter:

$$H_r(j\Omega) = \begin{cases} 1 & -\pi/T < \Omega \leq \pi/T \\ 0 & \text{otherwise} \end{cases} \quad h_r(t) = \frac{\sin(\pi t/T)}{(\pi t/T)}$$



Is $x_r(t) = x_c(t)$?

Proof in (1) frequency domain (clear!),

in (2) time domain?

(A) Check $t = nT$

$$h_r(t) = \begin{cases} 1, & n = 0 \\ 0, & n = \pm 1, \pm 2, \dots \end{cases}$$

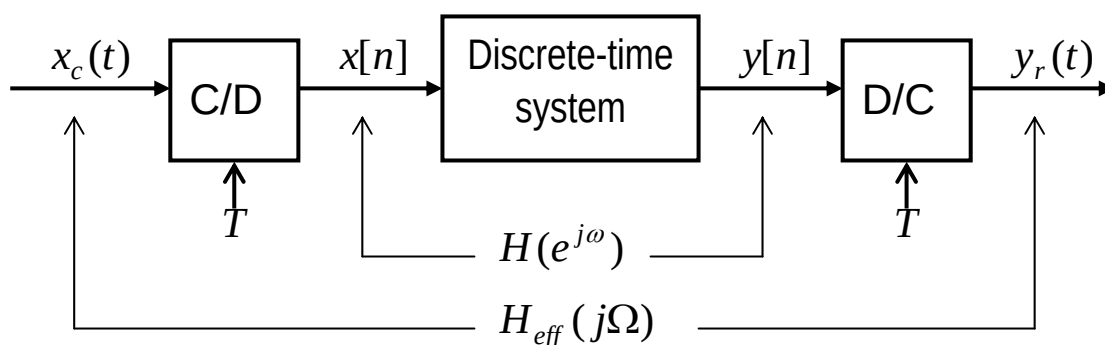
$$\rightarrow x_r(t) = x[n] = x_c(t)$$

(B) Check $kT < t < (k+1)T$ (Fig.4.8 on p.194)

$$x_r(t) = \sum x[n] \frac{\sin[\pi(t - nT)/T]}{\pi(t - nT)/T}$$

This is an interpolation – a weighted sum of $x[n]$'s. Note that in general we need infinite terms to reproduce $x_c(t)$. In practice, only the near-by terms matter. (We will later discuss the zero-order hold interpolator.)

4.4 Discrete-Time Processing of Continuous-Time Signals



- If this is an LTI system,

$$(1) \quad x[n] \rightarrow y[n]: Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

$$(2) \quad x_c(t) \rightarrow x[n]: \quad X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$$

$$(3) \quad y[n] \rightarrow y_r(t): \quad Y_r(j\Omega) = H_r(j\Omega)Y(e^{j\Omega T})$$

Note that “ T ” is included in the expression of $Y(e^{j\omega})$,

$\omega \leftarrow \Omega T$. This means “physical” frequency (not normalized).

$$(4) \quad x_c(t) \rightarrow \cdots \rightarrow y_r(t):$$

$$\begin{aligned} Y_r(j\Omega) &= H_r(j\Omega)H(e^{j\Omega T})X(e^{j\Omega T}) \\ &= H_r(j\Omega)H(e^{j\Omega T})\frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\Omega - j\frac{2\pi k}{T}\right) \end{aligned}$$

If $H_r(j\Omega)$ is an ideal low-pass filter, then

$$Y_r(j\Omega) = \begin{cases} H(e^{j\Omega T})X_c(j\Omega), & |\Omega| < \pi/T \\ 0, & \text{otherwise} \end{cases}$$

In other words, if $x_c(t)$ is *bandlimited* and is *ideally sampled* at a rate above the *Nyquist rate*, and the reconstruction filter is the *ideal low-pass* filter, then the *equivalent analog* filter has the same spectrum shape of the discrete-time filter.

$$H_{eff}(j\Omega) = \begin{cases} H(e^{j\Omega T}), & |\Omega| < \pi/T \\ 0, & \text{otherwise} \end{cases}$$

Remark: In order to have the above equivalent relation between $H(e^{j\omega})$ and $H_{eff}(j\Omega)$, we need

- (i) The system is LTI;
- (ii) The input is bandlimited;
- (iii) The input is sampled without aliasing and the ideal impulse train is used in sampling;
- (iv) The ideal reconstruction filter is used to produce the analog output.

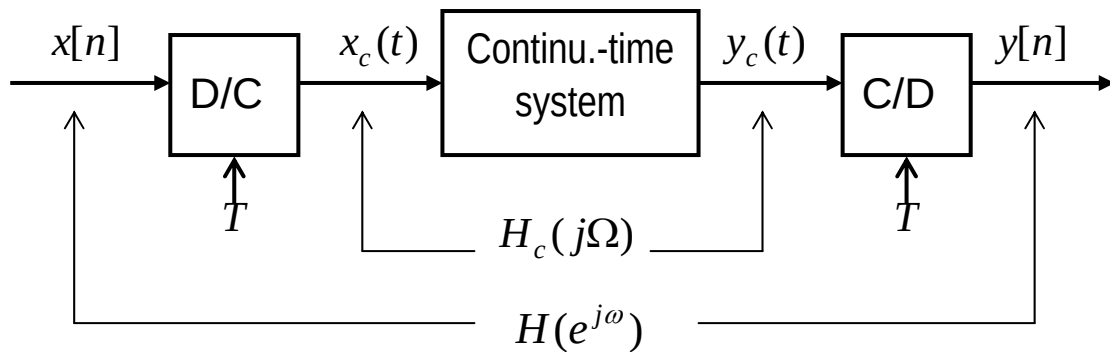
In practice, the above conditions are only approximately valid at best.

However, there are methods in designing the sampling and the reconstruction processes to make the approximation better.

- **Impulse invariance**

A filter design method – design a discrete-time filter that approximates an analog filter. We simply sample the analog filter impulse response $h[n] = Th_c(nT)$. (Details will be discussed in the filter design chapter.)

4.5 Continuous-Time Processing of Discrete-Time Signals



$$X_c(j\Omega) = TX(e^{j\Omega T}), \quad |\Omega| < \frac{\pi}{T}$$

$$Y_c(j\Omega) = H_c(j\Omega)X_c(j\Omega), \quad |\Omega| < \frac{\pi}{T}$$

$$Y(e^{j\omega}) = \frac{1}{T}Y_c(j\frac{\omega}{T}), \quad |\omega| < \pi$$

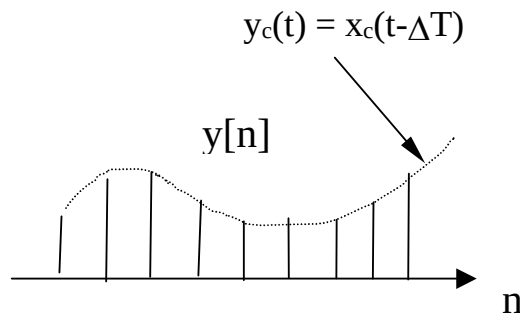
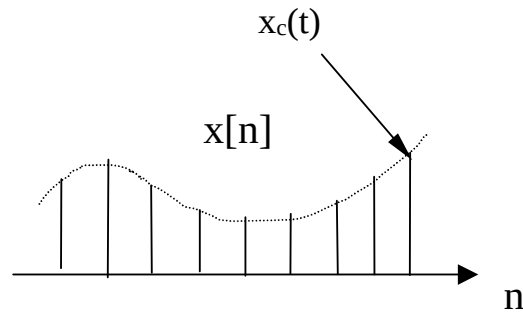
$$\Rightarrow Y(e^{j\omega}) = \frac{1}{T} H_c(j\frac{\omega}{T}) X_c(j\frac{\omega}{T}) = H_c(j\frac{\omega}{T}) X(e^{j\omega}) \quad |\omega| < \pi$$

$$\Rightarrow H(e^{j\omega}) = H_c(j\frac{\omega}{T}) \quad |\omega| < \pi$$

$$\text{or, equivalently,} \quad H(e^{j\Omega T}) = H_c(j\Omega) \quad |\Omega| < \frac{\pi}{T}$$

Example: Noninteger Delay

$$H(e^{j\omega}) = e^{-j\omega\Delta} \quad |\omega| < \pi$$



$$\begin{aligned} y[n] &= y_c(nT) = x_c(nT - \Delta T) \\ &= \sum_{k=-\infty}^{\infty} x[k] \frac{\sin[\pi(t - \Delta T - kT)/T]}{\pi(t - \Delta T - kT)/T} \Big|_{t=nT} \\ &= \sum_{k=-\infty}^{\infty} x[k] \frac{\sin[\pi(n - k - \Delta)]}{\pi(n - k - \Delta)} \end{aligned}$$

4.6 Change the Sampling Rate Using Discrete-time Processing

$$x_c(t) \begin{cases} \rightarrow T \rightarrow & x[n] = x_c(nT) \\ \rightarrow T' \rightarrow & x'[n] = x_c(nT') \end{cases}$$

Original sampling period: T

New sampling period: T'

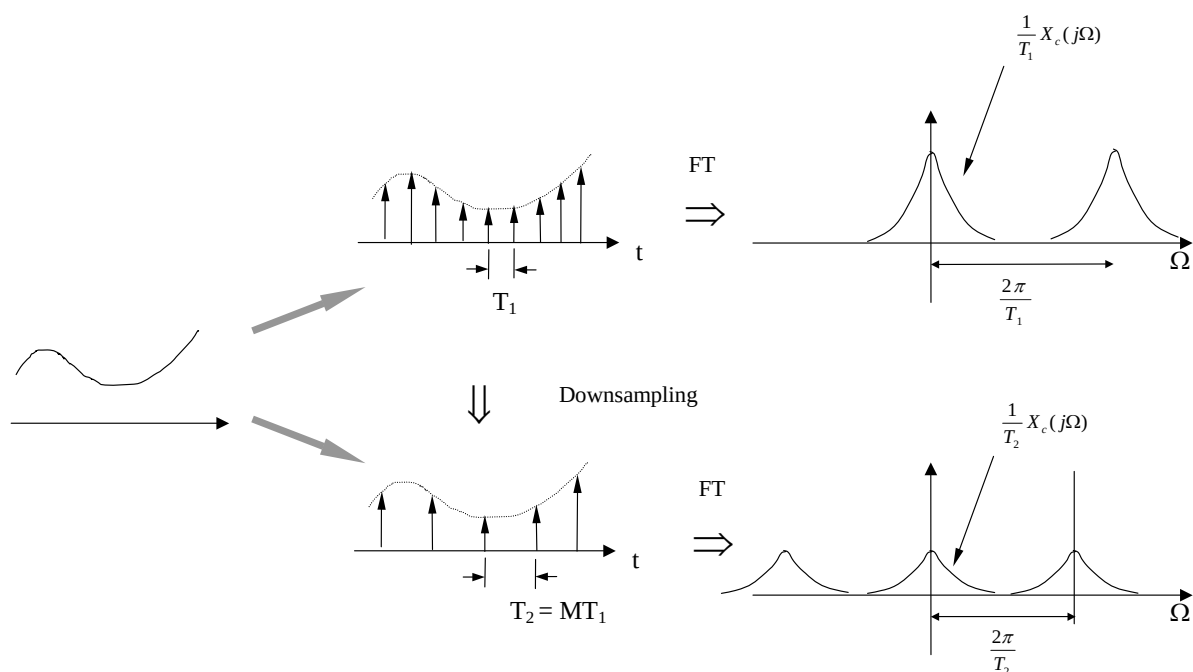
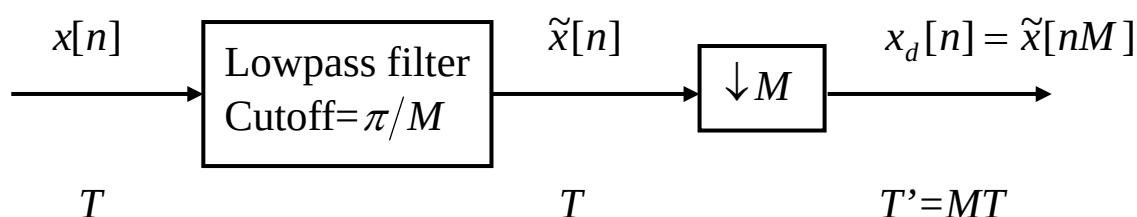
$$T \neq T'$$

How to do it if we don't go back to the continuous-time signal?

- **Sampling rate reduction by an integer factor**

■ **Sampling rate compressor:** $T' = MT$, where M is an integer

$$x_d[n] = x[nM] = x_c(nMT)$$



- (1) Aliasing: If the original signal BW is not small enough to meet the Nyquist rate requirement, prefiltering is needed.
- (2) Downsampling: The time-domain equation is simple as shown in the above. How does it look like in *frequency domain*?

The Original**The Downsampled**

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c \left(j \left(\frac{\omega}{T} - \frac{2\pi k}{T} \right) \right)$$

$$X_d(e^{j\omega}) = \frac{1}{T'} \sum_{r=-\infty}^{\infty} X_c \left(j \left(\frac{\omega}{T'} - \frac{2\pi r}{T'} \right) \right)$$

r	$\dots$	-2	-1	0	1	2	$\dots$	$(M-1)$	M	$(M+1)$	$\dots$
i	$\dots$	$(M-2)$	$(M-1)$	0	1	2	$\dots$	$(M-1)$	1	2	$\dots$
k	$\dots$	-1	-1	0	0	0	$\dots$	0	1	1	$\dots$

Old and new index: $r = i + kM$

$$r = -\infty, \dots, -2, -1, 0, 1, 2, \dots, \infty$$

$$k = -\infty, \dots, -2, -1, 0, 1, 2, \dots, \infty$$

$$i = 0, 1, 2, \dots, M-1$$

$$\begin{aligned}
 X_d(e^{j\omega}) &= \frac{1}{T'} \sum_{r=-\infty}^{\infty} X_c \left(j \left(\frac{\omega}{T'} - \frac{2\pi r}{T'} \right) \right) \\
 &= \frac{1}{MT} \sum_{r=-\infty}^{\infty} X_c \left(j \left(\frac{\omega}{MT} - \frac{2\pi r}{MT} \right) \right) \\
 &= \frac{1}{MT} \sum_{k=-\infty}^{\infty} \sum_{i=0}^{M-1} X_c \left(j \left(\frac{\omega}{MT} - \frac{2\pi kM}{MT} - \frac{2\pi i}{MT} \right) \right) \\
 X_d(e^{j\omega}) &= \frac{1}{M} \sum_{i=0}^{M-1} \left[\frac{1}{T} \sum_{k=-\infty}^{\infty} X_c \left(j \frac{\omega - 2\pi i}{MT} - j \frac{2\pi k}{T} \right) \right] \\
 &= \frac{1}{M} \sum_{i=0}^{M-1} X \left(e^{j \left(\frac{\omega}{M} - \frac{2\pi i}{M} \right)} \right)
 \end{aligned}$$

The downsampled spectrum = sum of shifted replica of the original
(Fig. 4.21: (1) shape expands; (2) replicas are inserted)

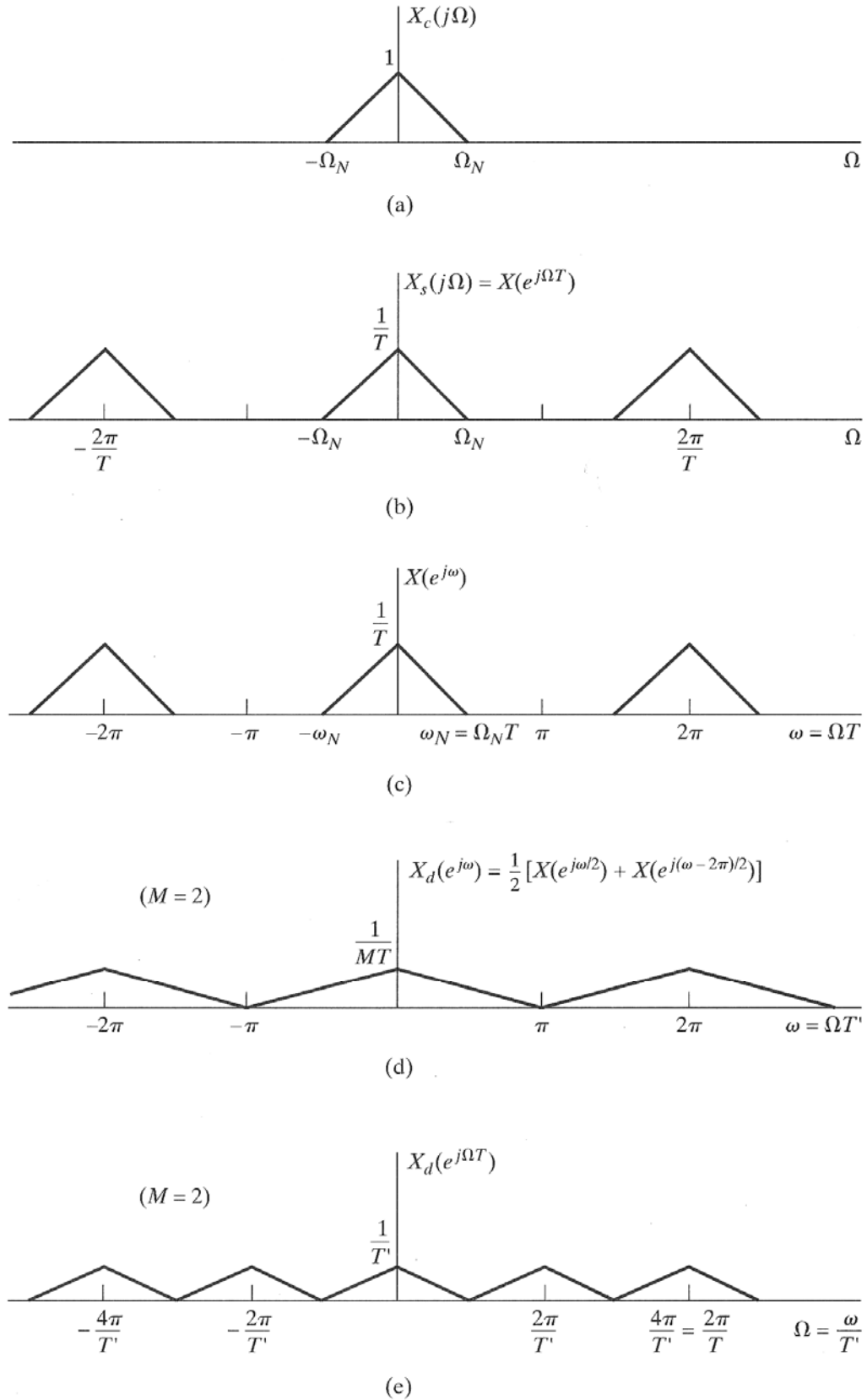


Figure 4.21 Frequency-domain illustration of downsampling.

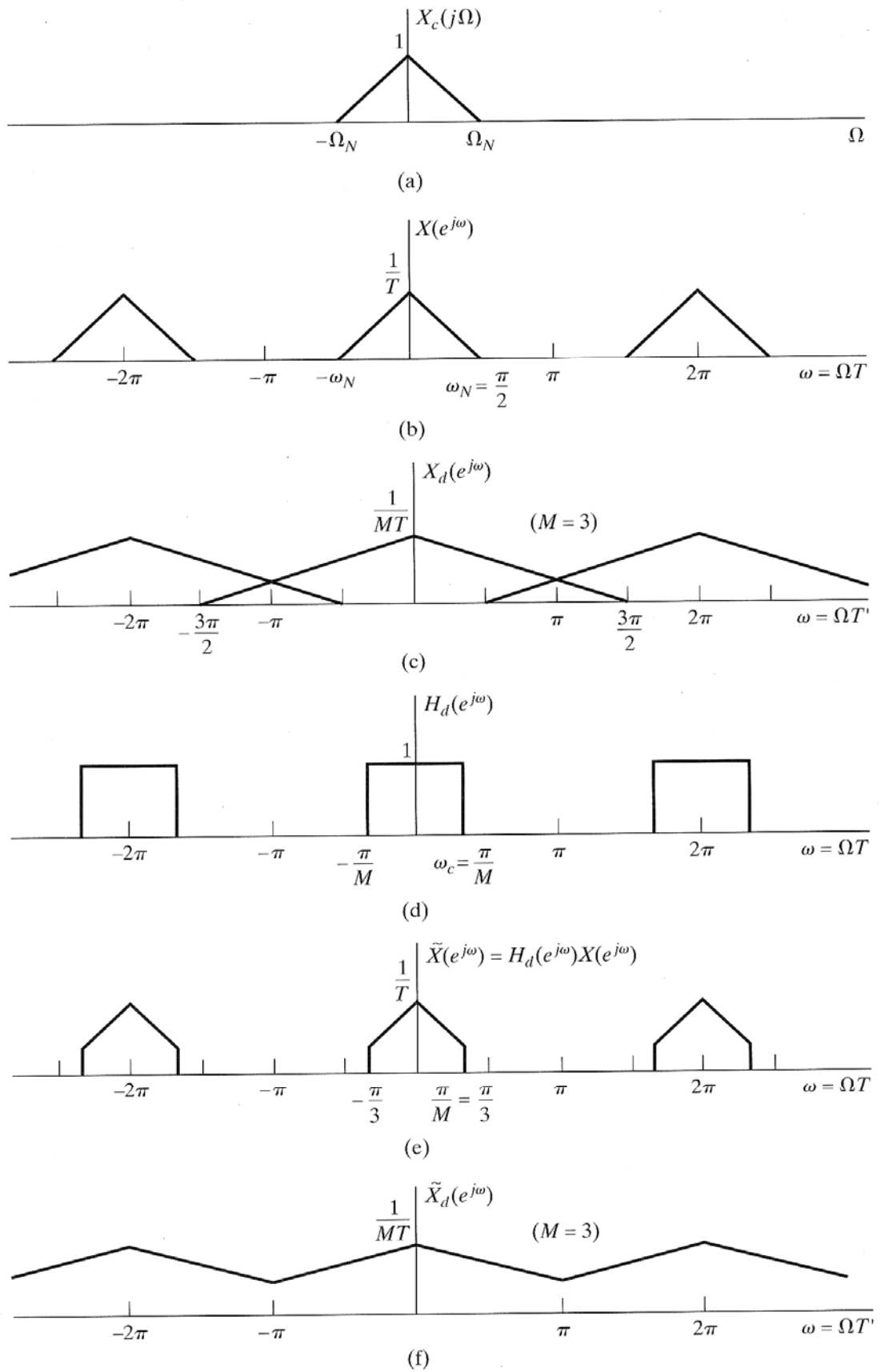
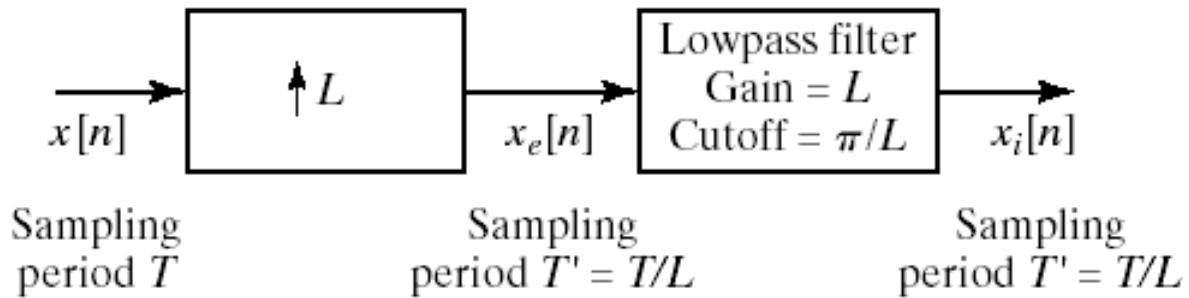


Figure 4.22 (a)–(c) Downsampling with aliasing. (d)–(f) Downsampling with prefiltering to avoid aliasing.

- **Increasing sampling rate by an integer factor**
 - **Sampling rate expander** $T' = T / L$, where L is an integer

$$x_i[n] = x_c\left(n \frac{T}{L}\right)$$



(Fig. 4.23: (1) shape is compressed; (2) replicas are removed)

(1) Increase samples

<Time-domain>

$$x_e[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases}$$

$$= \sum_{k=-\infty}^{\infty} x[k] \delta[n - kL]$$

<Frequency-domain>

$$X_e(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x[k] \delta[n - kL] \right) e^{-j\omega n}$$

$$= \sum_{k=-\infty}^{\infty} x[k] \left(\sum_{n=-\infty}^{\infty} \delta[n - kL] e^{-j\omega n} \right) = X(e^{j\omega L})$$

Note that $\sum_{n=-\infty}^{\infty} \delta[n - kL] e^{-j\omega n} = e^{-j\omega Lk}$

Remark: Essentially, the horizontal frequency axis is compressed.

The shape of the spectrum is not changed. $old_ \omega = \Omega T$,

$$new_ \omega = \Omega T' = \Omega T / L, \quad old_ \omega = new_ \omega \cdot L$$

Remark: At this point, we only insert zeros into the original signal.

In time domain, this signal doesn't look like the original.

(2) Ideal lowpass filtering

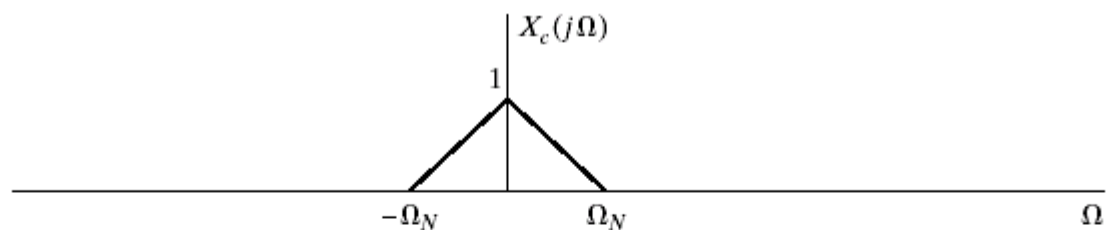
<Frequency-domain>

$$H_i(j\Omega) = \begin{cases} L & -\pi/(TL) < \Omega \leq \pi/(TL) \\ 0 & otherwise \end{cases}$$

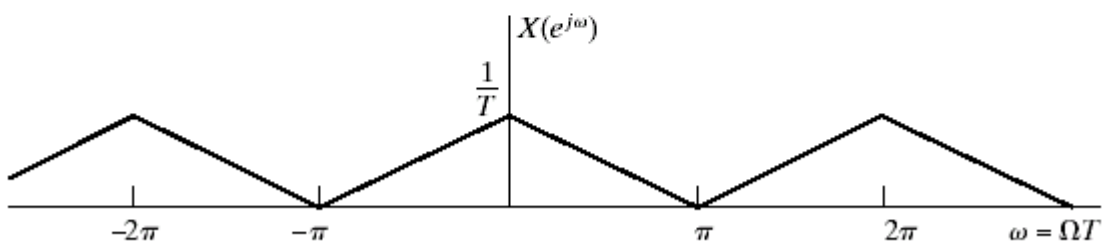
<Time-domain> (Fig. 4.26, insert "new" samples)

$$h_i[n] = \frac{\sin(\pi n/L)}{(\pi n/L)}, \text{ an interpolator!}$$

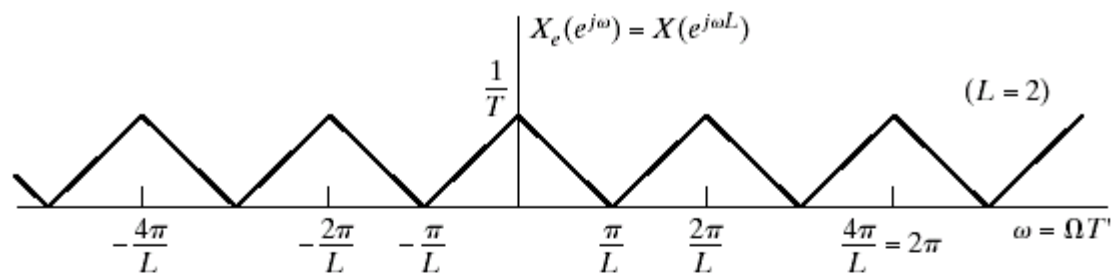
$$x_i[n] = \sum_{k=-\infty}^{\infty} x[k] \frac{\sin[\pi(n - kL)/L]}{\pi(n - kL)/L}$$



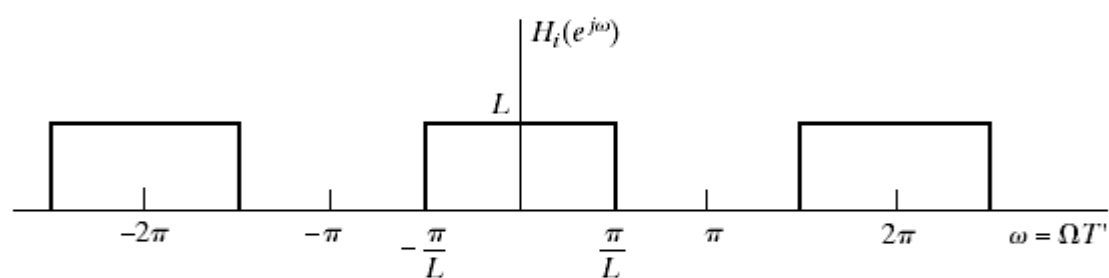
(a)



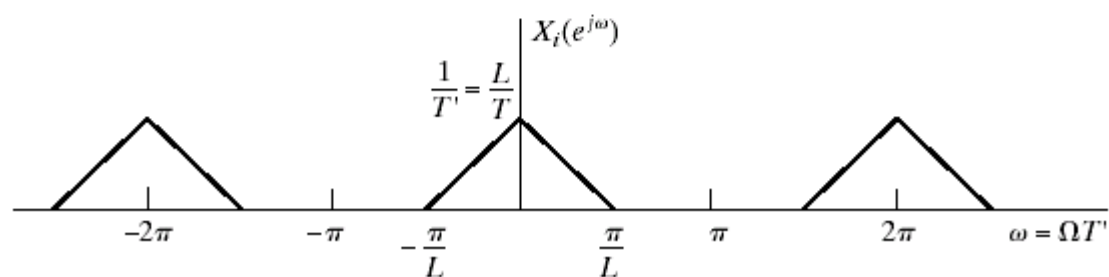
(b)



(c)



(d)



(e)

■ Linear interpolation

$$h_{lin}[n] = \begin{cases} 1 - |n|/L, & |n| \leq L \\ 0, & \text{otherwise} \end{cases}$$

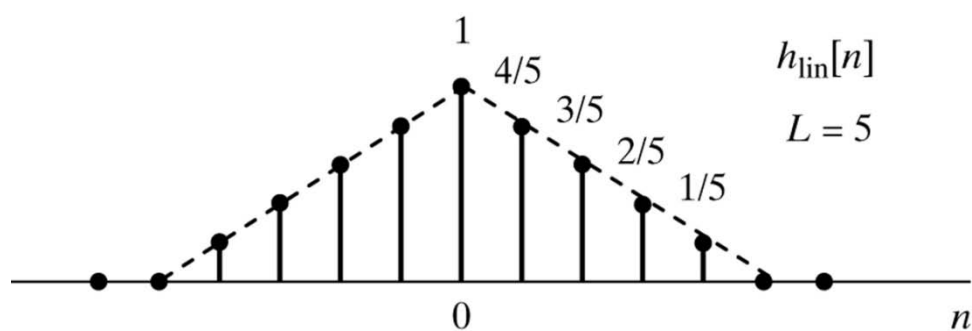
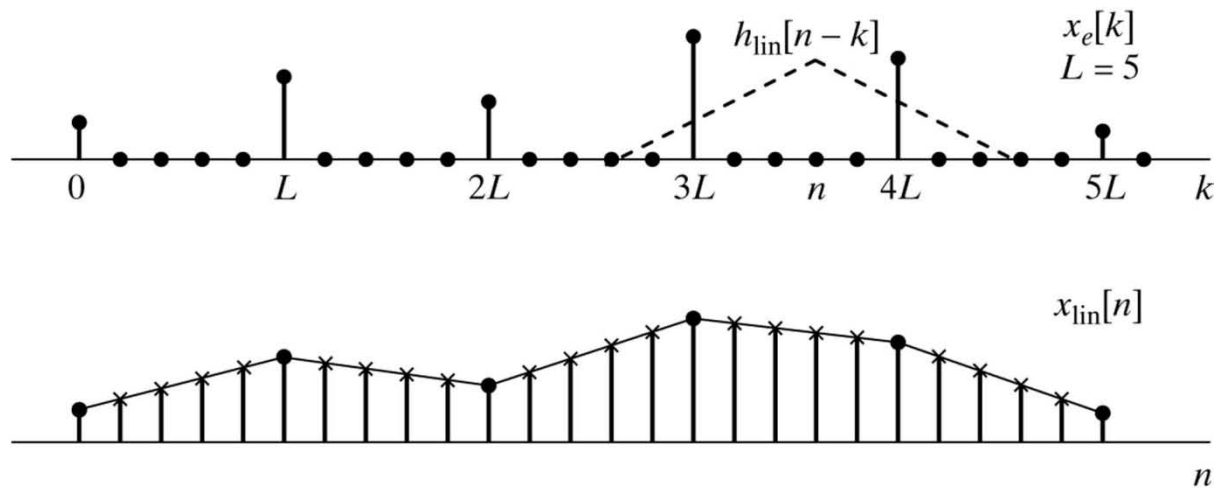


Fig. 4.25

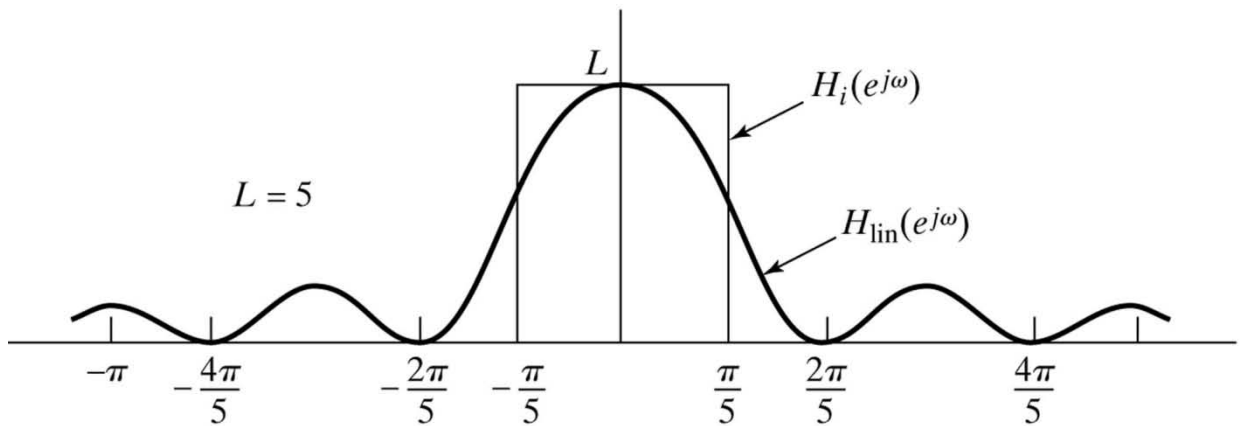
$$H_{lin}(e^{j\omega}) = \frac{1}{L} \left[\frac{\sin(\omega L/2)}{\sin(\omega/2)} \right]^2$$

$$x_{lin}[n] = \sum_{k=-\infty}^{\infty} x[k] h_{lin}[n - kL]$$

(Fig. 4.26, linear interpolation with $L=5$)



(a)



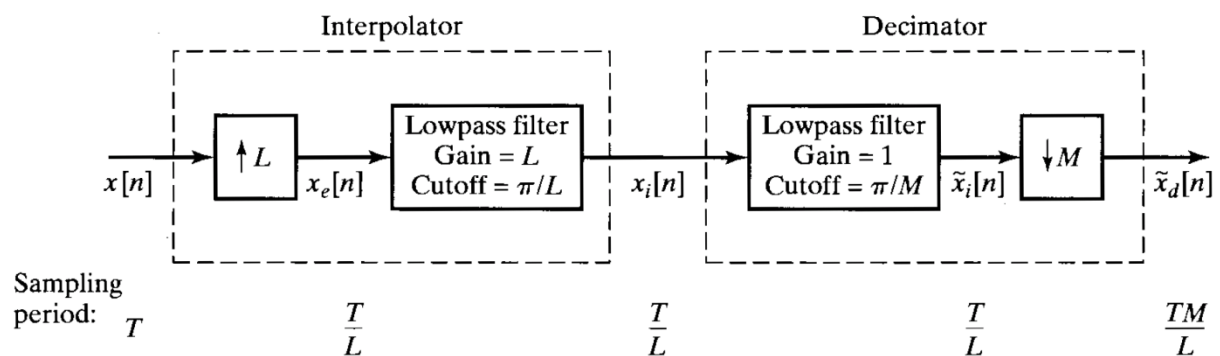
(b)

Figure 4.26 (a) Illustration of linear interpolation by filtering. (b) Frequency response of linear interpolator compared with ideal lowpass interpolation filter.

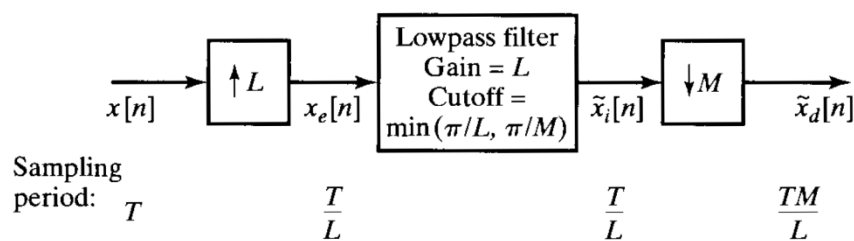
- Changing sampling rate by a rational factor

Idea: Sampling period $T \rightarrow \frac{M}{L}T$

$$T \xrightarrow{\text{interpolation}} \frac{T}{L} \xrightarrow{\text{decimation}} \frac{M}{L}T$$



(a)



(b)

Figure 4.29 (a) System for changing the sampling rate by a noninteger factor. (b) Simplified system in which the decimation and interpolation filters are combined.

Remark: In general, if the factor is not rational, go back to the continuous signals.

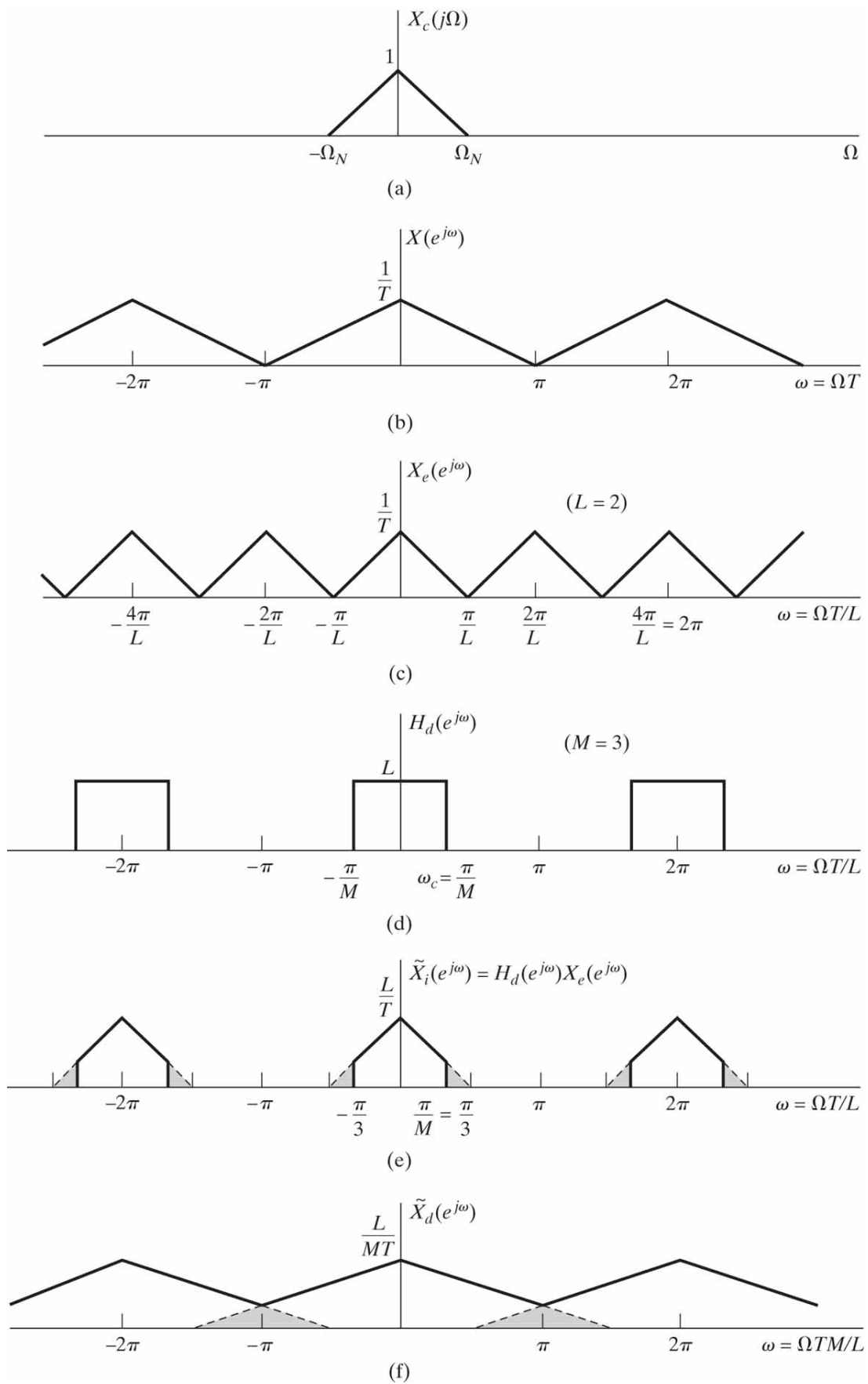


Figure 4.30 Illustration of changing the sampling rate by a noninteger factor.

- In summary:

-- Sampling

Time-domain	Frequency -domain
Prefiltering	Limit bandwidth $\Omega_s > 2\Omega_N$
Analog sampling (impulse train)	Duplicate and shift (Ω)
Analog to discrete $\delta(t) \rightarrow \delta[n]$	$\Omega \rightarrow \omega$

-- Reconstruction

Time-domain	Frequency -domain
Discrete to analog $\delta[n] \rightarrow \delta(t)$	$\omega \rightarrow \Omega$
Interpolation	Remove extra copies (Ω)

-- Down-sampling

Time-domain	Frequency -domain
Prefiltering	Limit bandwidth
Drop samples (rearrange index)	Expand (by a factor of M) and duplicate (insert (M-1) copies)

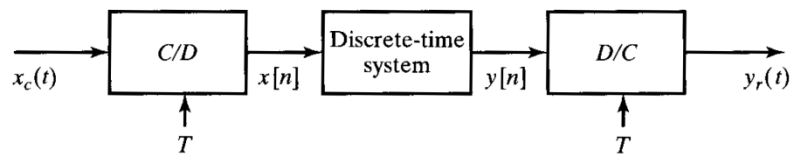
-- Up-sampling

Time-domain	Frequency -domain
Insert zeros	Shrink (by a factor of L)
Interpolation	Remove extra copies in a 2π period

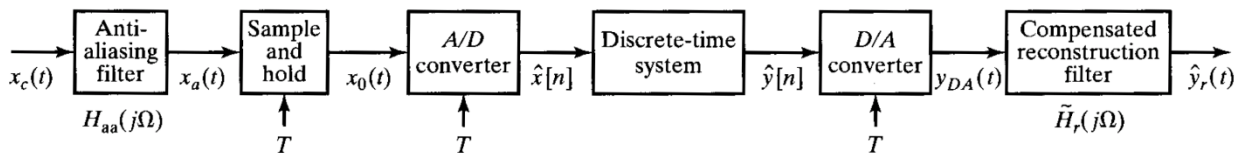
4.7 Digital Processing of Analog Signals

Ideal C/D converter $\rightarrow$ (approximation) analog-to-digital (A/D) converter

Ideal D/C converter $\rightarrow$ (approximation) digital-to-analog (D/A) converter



(a)



(b)

● Prefiltering to Avoid Aliasing

Ideal antialiasing filter: Ideal low-pass filter (difficult to implement sharp-cutoff analog filters).

◀ A solution: simple prefilter and oversampling followed by sharp antialiasing filters in discrete-time domain (Fig. 4.50)

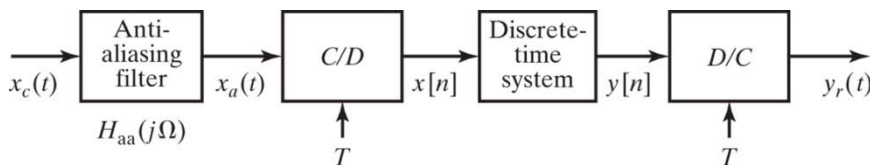
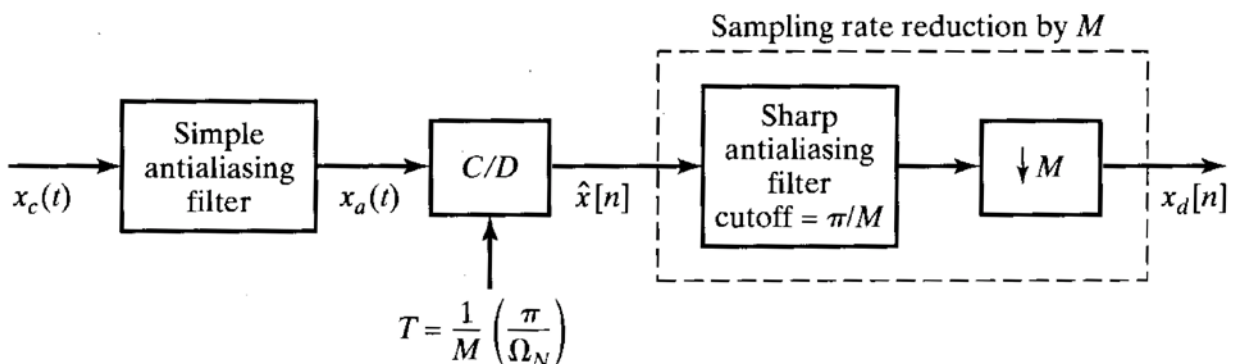


Fig. 4.48



- **A/D Conversion**

Digital: discrete in *time* and discrete in *amplitude*

Ideal sample-and-hold: (Fig. 4.46) Sample the (input) analog signal and hold its value for T seconds.

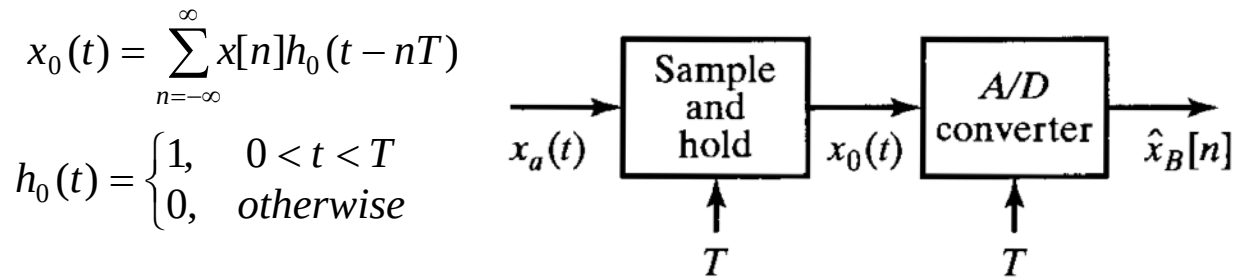
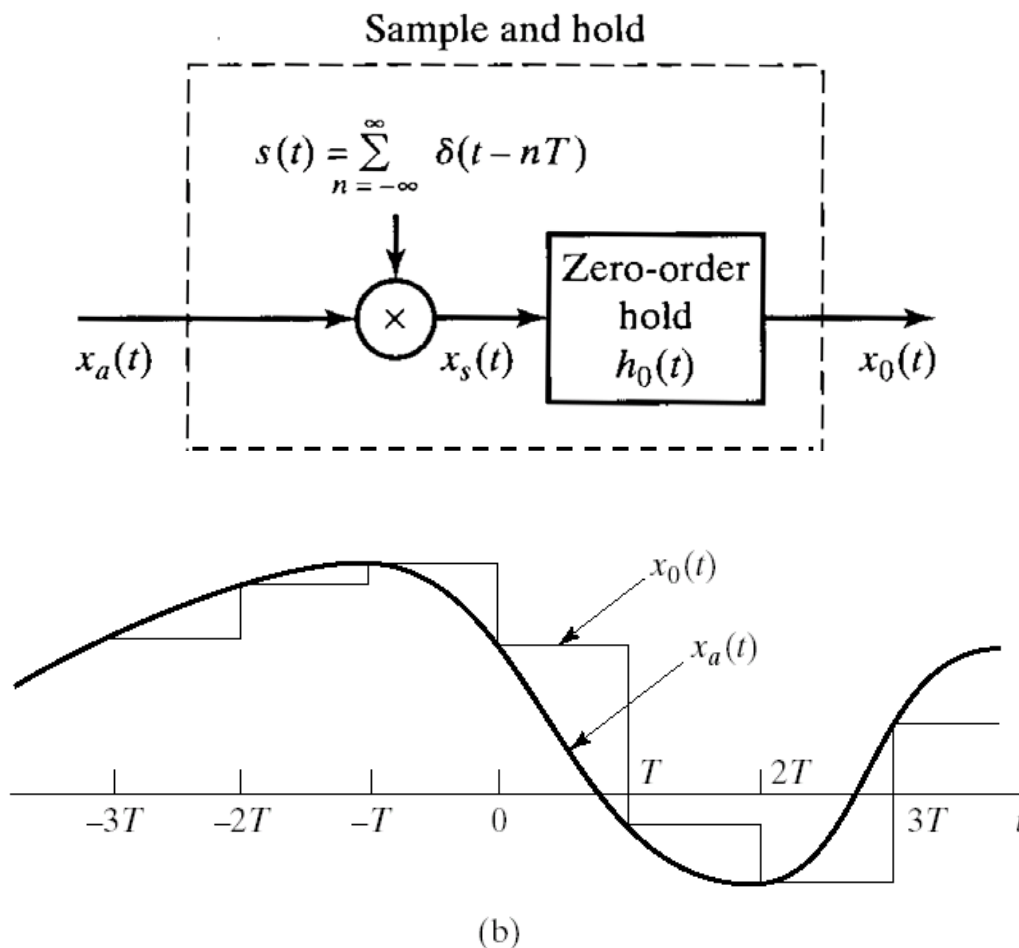


Fig. 4.51



Quantization: (Fig. 4.53) Transform the input sample $x[n]$ into one of a finite set of prescribed values.

$\hat{x}[n] = Q(x[n])$, $\hat{x}[n]$ is the quantized sample

Note: Quantization is a non-linear operation.

It is clear that $-\frac{\Delta}{2} < e[n] < \frac{\Delta}{2}$.

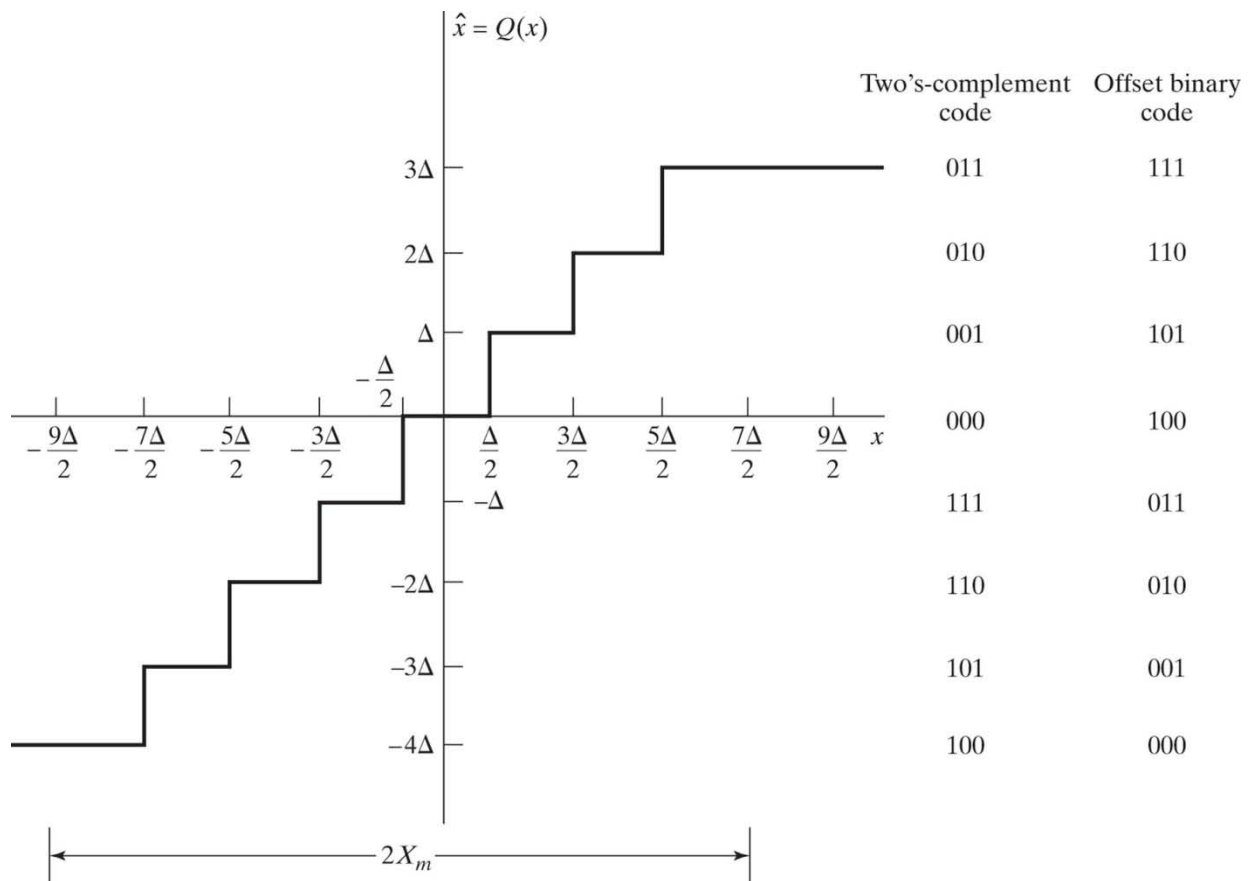


Figure 4.54 Typical quantizer for A/D conversion.

Statistical characteristics of $e[n]$:

- (1) $e[n]$ is stationary (probability distribution unchanged)
- (2) $e[n]$ is uncorrelated with $x[n]$
- (3) $e[n]$, $e[n+1]$, ... are uncorrelated (white)
- (4) $e[n]$ has a uniform distribution

The preceding assumptions are (approximately) valid if the signal is sufficiently *complex* and the quantization steps are sufficiently *small*, ...

Mean square error (MSE) of $e[n]$ (= variance if zero mean)

$$(\sigma_e)^2 = E\{(e - \bar{e})^2\} = \int_{-\Delta/2}^{\Delta/2} e^2 \frac{1}{\Delta} de = \frac{\Delta^2}{12}$$

-- Expressed in terms of 2^B and X_m

$$\sigma_e^2 = \frac{2^{-2B} X_m^2}{12}$$

-- SNR (signal-to-noise ratio) due to quantization

$$SNR = 10 \log_{10} \frac{\sigma_x^2}{\sigma_e^2} = 10 \log_{10} \frac{12 \cdot 2^{2B} \sigma_x^2}{X_m^2} = 10.8 + 6.02B - 20 \log_{10} \frac{X_m}{\sigma_x}$$

Remarks: (1) One bit buys a 6dB SNR improvement.

(2) If the input is Gaussian, a small percentage of the input samples would have an amplitude greater than $4\sigma_x$. If we choose

$$X_m = 4\sigma_x, \quad SNR \approx 6B - 1.25dB$$

For example, a 96dB SNR requires a 16-bit quantizer.

● D/A Conversion

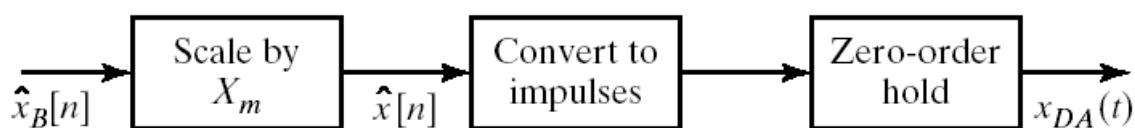
The ideal lowpass filter is replaced by “practical” filters.

Examples of practical filters: *zero-order hold* and *first-order hold*.

Mathematical model: (Fig. 4.62)



(a)



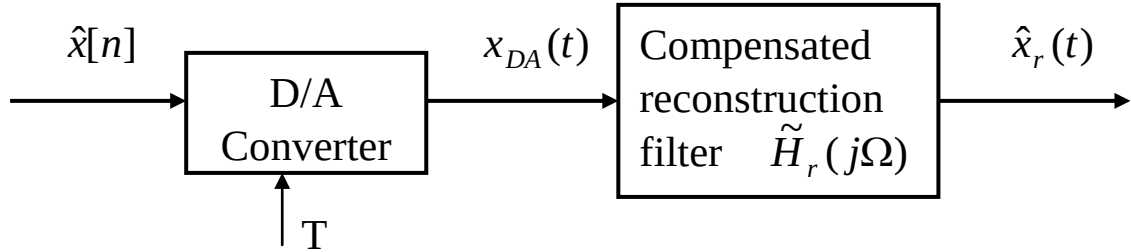
(b)

$$x_{DA}(t) = \sum_{n=-\infty}^{\infty} \hat{x}[n] h_0(t - nT)$$

= quantized input * impulse response of “practical” interpolation filter

$$\begin{aligned} x_{DA}(t) &= \sum_{n=-\infty}^{\infty} x[n] h_0(t - nT) + \sum_{n=-\infty}^{\infty} e[n] h_0(t - nT) \\ &= x_0(t) + e_0(t) \end{aligned}$$

Purpose: Find a compensation filter $\tilde{h}_r(t)$ to compensate for the distortion caused by the non-ideal $h_0(t)$ so that its output $\hat{x}_r(t)$ is close to the analog original $x_a(t)$.



In frequency domain:

$$\begin{aligned}
 X_0(j\Omega) &= F_t \left\{ \sum_{n=-\infty}^{\infty} x[n] h_0(t - nT) \right\} = \sum_{n=-\infty}^{\infty} x[n] H_0(j\Omega) e^{-j\Omega nT} \\
 &= \left(\sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega nT} \right) H_0(j\Omega) = X(e^{j\Omega T}) H_0(j\Omega)
 \end{aligned}$$

Because
$$X(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a \left(j \left(\Omega - \frac{2\pi k}{T} \right) \right),$$

$$X_0(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a \left(j \left(\Omega - \frac{2\pi k}{T} \right) \right) H_0(j\Omega)$$

[The interpolation filter $H_0(j\Omega)$ is used to remove the replicas.]

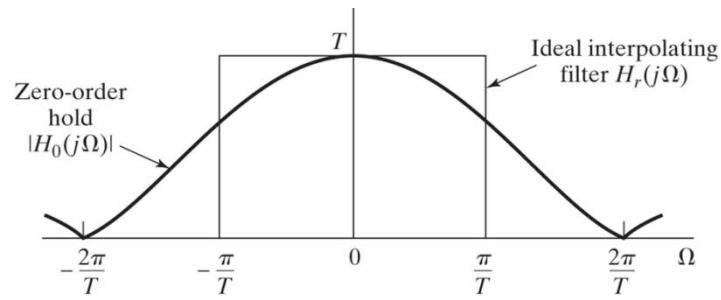
If $H_0(j\Omega)$ is not an ideal lowpass filter, we design a compensated reconstruction filter,

$$\tilde{H}_r(j\Omega) = \frac{H_r(j\Omega)}{H_0(j\Omega)}, \quad \text{where } H_r(j\Omega) \text{ is the ideal lowpass filter.}$$

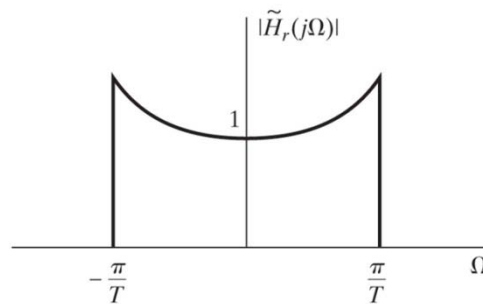
(1) **Zero-order hold**

$$h_0(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{otherwise} \end{cases} \quad \text{Or, } H_0(j\Omega) = \frac{2 \sin(\Omega T / 2)}{\Omega} e^{-j\Omega T / 2}$$

Thus, the compensated reconstruction filter is (Fig. 4.63)



(a)

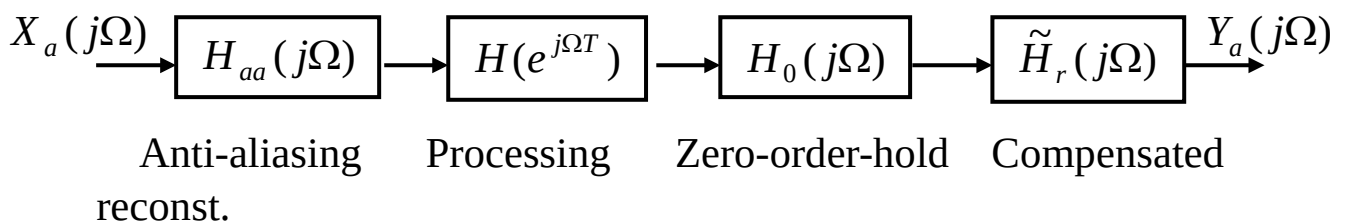


(b)

$$\tilde{H}_r(j\Omega) = \begin{cases} \frac{\Omega T / 2}{\sin(\Omega T / 2)} e^{j\Omega T / 2}, & |\Omega| < \pi / T \\ 0, & |\Omega| > \pi / T \end{cases}$$

Remark: A “practical” filter cannot achieve this $\leftarrow$ approximation.

Overall system:



$$Y_a(j\Omega) = \tilde{H}_r(j\Omega) \cdot H_0(j\Omega) \cdot H(e^{j\Omega T}) \cdot H_{aa}(j\Omega) X_a(j\Omega)$$

(2) **First-order hold**

Better frequency response than zero-order-hold. Predict the next $x(t)$ using the current sample $x(nT)$ and the slope between $x(nT-T)$ and $x(nT)$. (P&M)

$$\hat{x}(t) = x(nT) + \frac{x(nT) - x(nT - T)}{T}(t - nT), \quad nT \leq t < (n+1)T$$

$$h_0(t) = \begin{cases} 1 + \frac{t}{T}, & 0 \leq t \leq T \\ 1 - \frac{t}{T}, & T \leq t < 2T \\ 0, & \text{otherwise} \end{cases}$$

$$H_0(j\Omega) = T(1 + \Omega^2 T^2)^{1/2} \left[\frac{2 \sin(\Omega T / 2)}{\Omega T} \right]^2 e^{j\Theta(\Omega)}$$

$$\text{and } \Theta(\Omega) = -\frac{\Omega T}{2} + \tan^{-1}(\Omega T)$$

